# Pierpaolo **Vendittelli**

AI Researcher · Computational Pathology · Medical Imaging

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## EXPERIENCE

**Postdoc Researcher** *Dec. 2024 – Present*
**Prinses Maxima Centre for Pediatric Oncology** *Utrecht, The Netherlands*

- Developing DL and MIL pipelines for WSI analysis of Wilms tumor (anaplasia grading, clinical outcome prediction) across multiple collaborating hospitals.
- Translating model outputs into clinically interpretable biomarkers in collaboration with pathologists; supervising students and contributing to publications.

**PhD Candidate – AI for Digital Pathology** *Mar. 2021 – Jun. 2026 (exp.)*
**Radboud University Medical Center** *Nijmegen, The Netherlands*

- Designed multi-class U-Net for automated TSR quantification (PANCAIM, Horizon 2020): 368 patients / 4 datasets / 25 NL labs; Dice 0.75–0.86, TSR AUC 0.61 on external TCGA-PAAD cohort. Published in *PLOS ONE* (2024) and *SPIE MI* (2022).
- Teaching Assistant (*Image Analysis*, *Medical Imaging Analysis*, 4 editions); co-supervised Master's theses.

**Machine Learning Engineer** *Oct. 2020 – Feb. 2021*
**Everis Italy (NTT DATA)** *Rome, Italy*

- Built ML solutions on GCP (Vertex AI, BigQuery ML) for clients in banking, fashion, and automotive.

**Research Intern** *Feb. 2020 – Sep. 2020*
**ViCOROB – NeuroImage Computing Group, University of Girona** *Girona, Spain*

- Benchmarked MRI synthesis architectures (ResU-Net, cGAN, CycleGAN) on WMH and BraTS 2018; ResU-Net achieved best SSIM/MSE under data-scarce conditions.

## PUBLICATIONS & PROJECTS

**Automatic quantification of tumor-stroma ratio as a prognostic biomarker for pancreatic cancer** *PLOS ONE*
**doi:10.1371/journal.pone.0301969** *May 2024*
Multi-class U-Net; 368 patients / 4 datasets / 25 NL labs; Dice 0.75–0.86; TSR AUC 0.61 (6-month survival, TCGA-PAAD external cohort).

**Automatic quantification of TSR as a prognostic marker for pancreatic cancer** *MIDL 2023*
**openreview.net** *Sep. 2023*

**Setting the research agenda for clinical AI in pancreatic adenocarcinoma imaging** *Cancers*
**doi:10.3390/cancers14143498** *Jul. 2022*
Systematic review on AI for pancreatic cancer diagnosis and personalised treatment.

**Automatic tumour segmentation in H&E-stained whole-slide images of the pancreas** *SPIE MI 2022*
**doi:10.1117/12.2611542** *Apr. 2022*

**Weekly AI Paper-Triage Agent** *Personal Project*
**github.com/PierpaoloV** *2024 – Present*
Multi-agent LLM pipeline (OpenAI API, Python): weekly literature scan → summarisation → interest scoring → email digest.

## EDUCATION

**Joint M.Sc. in Medical Imaging and Applications (MAIA)** *Sep. 2018 – Sep. 2020*
**University of Burgundy, University of Cassino, University of Girona** *France, Italy, Spain*

- Grades: Bien (FR), 102/110 (IT), 8.6/10 (ES) • Erasmus Mundus Consortium Grant

**B.Sc. in Informatics and Telecommunication Engineering** *Sep. 2012 – Sep. 2017*
**University of Cassino and Southern Lazio** *Cassino, Italy*

- Grade: 94/110 • Erasmus+ Exchange, Fontys University of Applied Sciences, Eindhoven (2016)

## SKILLS

**Languages**: Python, C++, C, Bash, SQL

**ML / Deep Learning**: PyTorch, scikit-learn, Hugging Face Transformers, MONAI, Elastix, Keras
**MLOps & Infrastructure**: Docker, Git, GCP (Vertex AI, BigQuery ML), SLURM / GPU clusters, OpenSlide, W&B
**LLM & Agents**: OpenAI API, Anthropic API, RAG, prompt engineering, multi-agent orchestration